Percentages

Problem-Solving and Data Analysis · Math · 84 questions

1 026410-DC · Easy

During a sale, the original prices of all the items in a clothing store have been reduced by 20%. What is the sale price of a jacket with an original price of $50 ?

A. $12

B. $30

C. $36

D. $40

2 08199-DC · Easy

Call Ratings

	1 Star	2 Stars	3 Stars	4 Stars	Total
Employee A	16	49	72	8	145
Employee B	4	10	22	34	70
Employee C	8	56	45	16	125
Employee D	22	42	84	12	160
Total	50	157	223	70	500

A supervisor at a call center reviewed 500 calls taken by four employees and rated the employees' performance on each call on a scale from 1 star to 4 stars. The ratings for each employee are shown in the table above. According to the table, to the nearest whole percent, what percent of Employee A's calls received a rating of 1 star?

A. 3%

B. 11%

C. 16%

D. 32%

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3 374a0f38-19f3-4644-8e99-cb97adaa27f3 · Easy

Of 900,000 beads, 828,000 are silver. What percentage of the beads are silver?

A. 8%

B. 36%

C. 72%

D. 92%

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4 14922-DC · Easy

Rita’s total bill at a restaurant was $25.00, including tax. If she left a tip of 20% of the total bill, what was the amount of the tip?

A. $3.50

B. $4.00

C. $4.50

D. $5.00

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5 10596-DC · Easy

Lani spent 15% of her 8-hour workday in meetings. How many minutes of her workday did she spend in meetings?

A. 1.2

B. 15

C. 48

D. 72

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6 828689ef-29d6-4a4b-9945-060543cb1150 · Easy

There are a total of 840 seats in a school auditorium. During an assembly, students occupied 50% of the seats in the auditorium. How many seats did the students occupy during this assembly?

A. 25

B. 50

C. 420

D. 790

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7 06469-DC · Easy

Where Do People Get Most of Their Medical Information?

Source	Percent of those surveyed
Doctor	63%
Internet	13%
Magazines/brochures	9%
Pharmacy	6%
Television	2%
Other/none of the above	7%

The table above shows a summary of 1,200 responses to a survey question. Based on the table, how many of those surveyed get most of their medical information from either a doctor or the Internet?

A. 865

B. 887

C. 912

D. 926

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8 d82d4647-67d1-4ebf-8750-cf2ccee38a60 · Easy

What is 10% of 470?

A. 37

B. 47

C. 423

D. 460

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9 ee20c3bc-a074-467b-a5ec-dafc0c828d64 · Easy

There are 350 objects in a box. Of these objects, 4% are balls. How many balls are in the box?

A. 4

B. 14

C. 70

D. 346

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10 028969-DC · Easy

What number is 20% greater than 60 ?

A. 50

B. 72

C. 75

D. 132

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11 058dbe26-68a9-4661-8276-5e77b75ac679 · Easy

Of 300,000 paper clips, 234,000 are size large. What percentage of the paper clips are size large?

A. 22%

B. 33%

C. 66%

D. 78%

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12 baded933-3db8-4cfd-9e63-74113b894ad5 · Easy

There are 170 blocks in a container. Of these blocks, 10% are green. How many blocks in the container are green?

Student-produced response: _____

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13 20029b3e-d1cb-42ea-9a4b-f926abe12665 · Easy

Last year, 200 students enrolled in an interior design program. This year, the number of students enrolled is 147% of last year's number. How many students are enrolled in the interior design program this year?

A. 247

B. 294

C. 347

D. 394

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14 029360-DC · Easy

There were approximately 113,000 occupational therapy jobs in the United States in 2012. The Bureau of Labor Statistics has projected that this number will increase by 29% from 2012 to 2022. Of the following, which is closest to the number of occupational therapy jobs the bureau has projected for the United States in 2022?

A. 115,900

B. 116,300

C. 142,000

D. 145,800

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15 691ec468-2184-424c-ac94-53b3d07a4270 · Easy

Isabel grows potatoes in her garden. This year, she harvested 760 potatoes and saved 10% of them to plant next year. How many of the harvested potatoes did Isabel save to plant next year?

A. 66

B. 76

C. 84

D. 86

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16 6c847302-0dac-4944-8362-0bed796f6728 · Easy

There are 320 marbles in a container. Of these marbles, 10% are red. How many marbles in the container are red?

A. 32

B. 288

C. 320

D. 352

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17 70496145-8af1-415b-9df1-cb281b9bc4d2 · Easy

The length of the base of a certain parallelogram is 89% of the height of the parallelogram. Which expression represents the length of the base of the parallelogram, where h is the height of the parallelogram?

A. $89h$

B. $0.089h$

C. $8.9h$

D. $0.89h$

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18 028060-DC · Easy

The cost of a certain shirt is $20 before a 5% sales tax is added. What is the total cost, including sales tax, to purchase the shirt?

A. $20.05

B. $20.50

C. $21.00

D. $25.00

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19 8ac6b4d3-1459-4464-8d25-73cd50d43d67 · Easy

What is 20% of 440?

A. 44

B. 88

C. 880

D. 1,760

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20 ae635e6f-331e-4b53-b528-468662017b4e · Easy

What is 10% of 370?

A. 27

B. 37

C. 333

D. 360

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21 fc5659d4-90ce-4136-b70e-ac4dd697c151 · Easy

21 is 21% of what number?

A. 0

B. 1

C. 42

D. 100

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22 d76b8468-eb1c-4309-9278-5eda6d2d9b3e · Easy

There are 250 trees in a park. Of these trees, 6% are birch trees. How many birch trees are in the park?

A. 6

B. 15

C. 75

D. 244

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23 46e791de-a8dd-4add-bf7d-3a1c1fbef6ce · Easy

Amara grows cherry tomatoes in her backyard. This year, she harvested 750 cherry tomatoes and gave 10% of them to her neighbor. How many of the harvested cherry tomatoes did Amara give to her neighbor?

Student-produced response: _____

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24 06907-DC · Easy

A waiter receives tips from each customer. On average, the tip is 15% of the customer's bill. At this rate, which of the following is closest to the tip the waiter can expect when a customer has a bill that is $78.20?

A. $8.00

B. $10.00

C. $12.00

D. $14.00

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25 f0fb6cfb-ea46-4a28-8379-6d0acc914299 · Easy

What is 23% of 100?

A. 23

B. 46

C. 77

D. 123

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26 070905-DC · Easy

A table of the US minimum wage for 6 different years is shown below.

Year	US minimum wage (dollars per hour)
1960	1.00
1970	1.60
1980	3.10
1990	3.80
2000	5.15
2010	7.25

What was the percent increase of the minimum wage from 1960 to 1970?

A. 30%

B. 60%

C. 62.5%

D. 120%

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27 e06a745d-b077-48ee-8763-fdfcd713b268 · Easy

What percentage of 300 is 75?

A. 25%

B. 50%

C. 75%

D. 225%

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28 71ad6fcc-0b8a-4040-a827-1cf9cc1ed910 · Easy

Out of 300 seeds that were planted, 80% sprouted. How many of these seeds sprouted?

Student-produced response: _____

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29 3b2762db-3e04-4423-acad-306a8ce4ed60 · Medium

What number is 40% greater than 115?

Student-produced response: _____

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30 0359b97d-4f74-4893-886b-a63458ac15d7 · Medium

Which expression represents the result of increasing a positive quantity w by 43%?

A. $1.43w$

B. $0.57w$

C. $43w$

D. $0.43w$

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31 b3f57376-4374-4b2d-ab34-98d991fed767 · Medium

The table summarizes the number of objects in each group.

Group	Number of objects
A	375
B	54
C	690
D	81
Total	1,200

The number of objects in group C is p% of the number of objects in group A. What is the value of p?

Student-produced response: _____

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32 026407-DC · Medium

Which of the following represents the result of increasing the quantity x by 9%, where $x > 0$?

A. $1.09x$

B. $0.09x$

C. $x + 9$

D. $x + 0.09$

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33 b90e5782-c428-473b-8c51-a587d2c330b9 · Medium

In a group, 40% of the items are red. Of all the red items in the group, 30% also have stripes. What percentage of the items in the group are red with stripes?

A. 10%

B. 12%

C. 70%

D. 75%

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34 031617-DC · Medium

The number k is 36% greater than 50. If k is the product of 50 and r, what is the value of r ?

A. 36

B. 3.6

C. 1.36

D. 0.36

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35 9d08821e-6f03-46fa-b7f1-5b1d8c3a6d3f · Medium

An insurance company offers a series of three information sessions. 1,250 people attended the first information session. 72% of the people who attended the first information session attended the second information session, and 36% of the people who attended the first and second information sessions attended the third information session. How many people attended all three information sessions?

Student-produced response: _____

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36 10b6d39c-f6c1-4d7f-a4e7-f4e5220bb807 · Medium

432 is 96% of what number?

Student-produced response: _____

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37 028999-DC · Medium

A number n is increased 6%. If the result is 318, what is the value of n ?

A. 199

B. 299

C. 300

D. 337

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38 db08d5bc-9768-4b87-8642-1033890325ea · Medium

The population of City A increased by 7% from 2015 to 2016. If the 2016 population is k times the 2015 population, what is the value of k?

A. 0.07

B. 0.7

C. 1.07

D. 1.7

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39 f518f607-0819-47bd-b33b-db922ee03cb2 · Medium

The value of z is 1.13 times 100. The value of z is what percent greater than 100?

A. 11.3

B. 13

C. 130

D. 213

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40 ccae50dc-b231-44c5-869e-aacfc573e043 · Medium

In a sample, 80% of the items are faulty. There are 88 faulty items in the sample. How many total items are in the sample?

Student-produced response: _____

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41 019667-DC · Medium

During the first month of sales, a company sold 1,300,000 units of a certain type of smartphone. During the same month, 15% of the units sold were returned. If sales and the return rate remain the same for each of the next 5 months, about how many units of this smartphone will be returned to the company during this 6-month period?

A. 195,000

B. 975,000

C. 1,170,000

D. 6,630,000

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42 05321-DC · Medium

Thomas installed a new stove in his restaurant. At the time of installation, the stove had a value of $800. Thomas estimates that each year the value of the stove will depreciate by 20% of the previous year's estimated value. What is the estimated value of the stove exactly 2 years after Thomas installed it?

A. $480

B. $512

C. $556

D. $640

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43 cb21253d-b0c2-4157-bf21-8fa351eb6b79 · Medium

The number of coins in a collection increased from 9 to 90. What was the percent increase in the number of coins in this collection?

A. 10%

B. 81%

C. 90%

D. 900%

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44 05323-DC · Medium

A customer's monthly water bill was $75.74. Due to a rate increase, her monthly bill is now $79.86. To the nearest tenth of a percent, by what percent did the amount of the customer's water bill increase?

A. 4.1%

B. 5.1%

C. 5.2%

D. 5.4%

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45 c58c16c7-0525-4a20-b6ce-33c59a2d682f · Medium

The population of Greenville increased by 7% from 2015 to 2016. If the 2016 population is k times the 2015 population, what is the value of k?

A. 0.07

B. 0.7

C. 1.07

D. 1.7

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46 967c754c-6ed6-4648-9b7b-4a17946cf25f · Medium

The amount of Hanna's bill for a food order was $\$ 50$. Hanna gave a tip of 20% of the amount of the bill. What is the amount, in dollars, of the tip Hanna gave?

Student-produced response: _____

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47 884dde13-7464-48c1-89f7-7a74bb1e025e · Medium

In March, the price of a collectible card was $\$ 15.50$. In April, the price of the collectible card was $\$ 17.36$. The price of the collectible card in April was $p\%$ of the price of the collectible card in March. What is the value of p?

A. 12

B. 88

C. 112

D. 188

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48 47d32f6a-d945-4d6e-9f0f-18a983b72be4 · Medium

Last year, Cedric had 35 plants in his garden. This year, the number of plants in Cedric's garden is 60% greater than the number of plants in his garden last year. How many plants does Cedric have in his garden this year?

Student-produced response: _____

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49 c709983b-f5d2-4f76-8755-5eb306b94baa · Medium

There are 450 tiles in a box. Of these tiles, 6% are black. How many black tiles are in the box?

Student-produced response: _____

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50 2ac7bdf3-7d6c-4145-a9a6-af9bf15e1765 · Medium

13 is $p\%$ of 25. What is the value of p?

Student-produced response: _____

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51 00142-DC · Hard

Year	Subscriptions sold
2012	5,600
2013	5,880

The manager of an online news service received the report above on the number of subscriptions sold by the service. The manager estimated that the percent increase from 2012 to 2013 would be double the percent increase from 2013 to 2014. How many subscriptions did the manager expect would be sold in 2014?

A. 6,020

B. 6,027

C. 6,440

D. 6,468

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52 65f570ee-64b0-4f20-8200-435cc6939ac8 · Hard

The expression $0.35x$ represents the result of decreasing a positive quantity x by what percent?

A. 3.5%

B. 35%

C. 6.5%

D. 65%

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53 8b00175d-6fbb-4911-bbd3-f2a1167fb3e5 · Hard

The number *a* is 190% greater than the number *b*. The number *b* is 80% less than 24. What is the value of *a*?

A. 9.12

B. 13.92

C. 26.40

D. 36.48

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54 9aa8fbdb-6e32-4b91-81d7-831d0518641e · Hard

After 20% of the original number of marbles in a group were removed from the group, 360 marbles remained in the group. How many marbles were removed from the group?

A. 72

B. 90

C. 450

D. 1,800

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55 a441304f-3293-46ff-a2c2-cdbb915b8b11 · Hard

The number *a* is 110% greater than the number *b*. The number *b* is 90% less than 47. What is the value of *a*?

Student-produced response: _____

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56 0e0b2da3-40d6-47ae-8c52-727851d9038b · Hard

A gift shop buys souvenirs at a wholesale price of 7.00 dollars each and resells them each at a retail price that is 290% of the wholesale price. At the end of the season, any remaining souvenirs are marked at a discounted price that is 80% off the retail price. What is the discounted price of each remaining souvenir, in dollars?

Student-produced response: _____

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57 ad721785-5e4c-4d71-bf8a-1acab4691b6c · Hard

The positive number a is $2{,}241\%$ of the sum of the positive numbers b and c, and b is 83% of c. What percent of b is a?

A. 23.24%

B. 49.41%

C. $2{,}324\%$

D. $4{,}941\%$

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58 22570c4d-ed8e-4040-ae8a-171921cd4f53 · Hard

At a recreation center, visitors used the exercise room for a total of t hours in July. At the same recreation center, visitors used the exercise room for $3.49t$ hours in August. What is the percent increase in the number of hours visitors used the exercise room from July to August?

A. 2.49%

B. 3.49%

C. 249%

D. 349%

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59 a643565e-80af-4950-9896-22d6389302c4 · Hard

The speeds of objects A, B, and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of object A is $5{,}900\%$ of the speed of object C and the speed of object C is 0.008% of the speed of object B, which expression represents the value of $a + b$ in terms of c?

A. $1{,}840c$

B. $5{,}908c$

C. $6{,}025c$

D. $12{,}559c$

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60 29c0769a-6466-4e2e-8793-1978ec02749c · Hard

The number *w* is 110% greater than the number *z*. The number *z* is 55% less than 50. What is the value of *w*?

Student-produced response: _____

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61 38d4bbcc-99b8-4fc3-8247-5add78d4c1dd · Hard

The number *a* is 60% greater than the positive number *b*. The number *c* is 45% less than *a*. The number *c* is how many times *b*?

Student-produced response: _____

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62 5cbbaf2c-1ba2-4545-b2ad-f26827975dbc · Hard

A square scale drawing has a side length of 55 inches, and 1 inch on the drawing represents an actual distance of 19 miles. A larger version of the same drawing is printed as a square with the side length 65% longer than the side length of the previous drawing. On the larger drawing, which of the following is closest to the actual distance, in miles, represented by 1 inch?

A. 6.65

B. 11.52

C. 31.35

D. 35.75

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63 dabe922d-3778-48a9-88e7-926763881b51 · Hard

37% of the items in a box are green. Of those, 37% are also rectangular. Of the green rectangular items, 42% are also metal. Which of the following is closest to the percentage of the items in the box that are not rectangular green metal items?

A. 1.16%

B. 57.50%

C. 94.25%

D. 98.84%

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64 989b4b44-3e81-4a7a-a64c-9ab381c1b983 · Hard

210 is *p*% greater than 30. What is the value of *p*?

Student-produced response: _____

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65 ee0361d9-4c33-463e-a8a8-6830e04fd051 · Hard

The regular price of a shirt at a store is $ 11.70. The sale price of the shirt is 80% less than the regular price, and the sale price is 30% greater than the store's cost for the shirt. What was the store's cost, in dollars, for the shirt? (Disregard the $ sign when entering your answer. For example, if your answer is $ 4.97, enter 4.97)

Student-produced response: _____

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66 5c387bc7-97b0-4bcf-b06a-892858d9b7b1 · Hard

The number of zebras in a population in 2018 was 1.27 times the number of zebras in this population in 2014. If the number of zebras in this population in 2014 is *p*% of the number of zebras in this population in 2018, what is the value of *p*, to the nearest whole number?

Student-produced response: _____

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67 101896fa-20f1-44ca-a5ec-1b422cb3fce3 · Hard

The number *a* is 55% less than the number *b*. The number *b* is 320% greater than 160. What is the value of *a*?

Student-produced response: _____

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68 04778-DC · Hard

A school district is forming a committee to discuss plans for the construction of a new high school. Of those invited to join the committee, 15% are parents of students, 45% are teachers from the current high school, 25% are school and district administrators, and the remaining 6 individuals are students. How many more teachers were invited to join the committee than school and district administrators?

Student-produced response: _____

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69 91c12a4d-8406-4ac0-bd8c-732595daf5a4 · Hard

The number *a* is 70% less than the positive number *b*. The number *c* is 80% greater than *a*. The number *c* is how many times *b*?

Student-produced response: _____

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70 1edfed1c-89c5-4702-a448-e023544a67b4 · Hard

A reseller buys certain books for a purchase price of 5.00 dollars each and then marks them each for sale at a consumer price that is 270% of the purchase price. After 4 months, any remaining books not yet sold are marked at a discounted price that is 70% off the consumer price. What is the discounted price of each of the remaining books, in dollars?

Student-produced response: _____

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71 36bef1c8-d853-462f-a9e8-7575812e84e9 · Hard

The weight of sample A is 468% of the weight of sample B, and the weight of sample A is 0.780% of the weight of sample C. If the weight of sample C is $p\%$ of the weight of sample B, what is the value of p?

Student-produced response: _____

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72 fb4b074b-8096-42a7-9ab5-1672ab4556c2 · Hard

The result of increasing the quantity x by 400% is 60. What is the value of x?

A. 12

B. 15

C. 240

D. 340

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73 f3910fb9-490b-407a-b587-704499bf653f · Hard

According to a set of standards, a certain type of substance can contain a maximum of 0.001% phosphorus by mass. If a sample of this substance has a mass of 140 grams, what is the maximum mass, in grams, of phosphorus the sample can contain to meet these standards?

Student-produced response: _____

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74 b52952eb-21e9-41d5-ac01-117586367fc5 · Hard

The number a is 70% less than the positive number b. The number c is 60% greater than a. The number c is how many times b?

Student-produced response: _____

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75 af7c40ef-acf4-4f14-9563-4a4d1a99a37d · Hard

The result of increasing the quantity x by $1{,}800\%$ is 684. What is the value of x?

A. $12{,}996$

B. $12{,}312$

C. 38

D. 36

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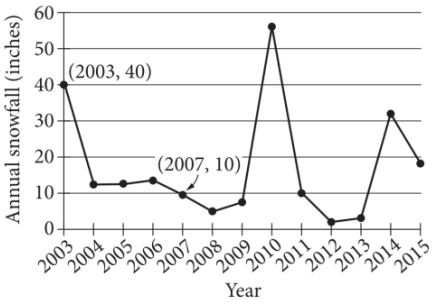
76 0eac6bca-d744-4e7a-be68-7d4922a300ce · Hard

The number of audiobooks in a library increased by 131% from last year to this year. The number of audiobooks in the library last year was n, and the number of audiobooks in the library this year is bn, where b and n are constants. What is the value of b?

Student-produced response: _____

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77 070908-DC · Hard



The line graph shows the total amount of snow, in inches, recorded each year in Washington, DC, from 2003 to 2015. If $p\%$ is the percent decrease in the annual snowfall from 2003 to 2007, what is the value of p ?

Student-produced response: _____

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78 070906-DC · Hard

Jennifer bought a box of Crunchy Grain cereal. The nutrition facts on the box state that a serving size of the cereal is $\frac{3}{4}$ cup and provides 210 calories, 50 of which are calories from fat. In addition, each serving of the cereal provides 180 milligrams of potassium, which is 5% of the daily allowance for adults. If p percent of an adult's daily allowance of potassium is provided by x servings of Crunchy Grain cereal per day, which of the following expresses p in terms of x ?

A. $p = 0.5x$

B. $p = 5x$

C. $p = (0.05)^x$

D. $p = (1.05)^x$

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79 fa595c60-481e-4838-966e-bb53dd5ec839 · Hard

The positive number *a* is 230% of the number *b*, and *a* is 60% of the number *c*. If *c* is *p*% of *b*, which of the following is closest to the value of *p*?

A. 138

B. 217

C. 283

D. 383

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80 71e0f3ea-a756-49f9-bc0d-dbe87bc5a198 · Hard

A scientist studying the life cycle of dragonflies counted the number of dragonflies in a certain habitat each day for 46 days. On February 15, there were 99 dragonflies in the habitat. The percent increase in the number of dragonflies in the habitat from January 1 to February 15 was 12.50%. How many dragonflies were in the habitat on January 1?

A. 88

B. 87

C. 12

D. 8

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81 58739008-cda2-4703-9c33-924423c8a7d7 · Hard

0.1152 is 0.08% of the number *b*. The number *b* is 25% of the number *c*. What is the value of *c*?

Student-produced response: _____

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82 02ff4aa9-be78-434c-a64d-d18b83b4e58a · Hard

The value of a collectible comic book increased by 167% from the end of 2011 to the end of 2012 and then decreased by 16% from the end of 2012 to the end of 2013. What was the net percentage increase in the value of the collectible comic book from the end of 2011 to the end of 2013?

A. 124.28%

B. 140.28%

C. 151.00%

D. 209.72%

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83 8edf9775-0d89-4ffa-8b53-27e4265b71f6 · Hard

140 is p% greater than 10. What is the value of p?

A. 1,400

B. 1,300

C. 140

D. 130

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84 bd6a0437-de79-407d-a6f8-ad2d753ff04e · Hard

In 2008, Zinah earned 14% more than in 2007, and in 2009 Zinah earned 4% more than in 2008. If Zinah earned y times as much in 2009 as in 2007, what is the value of y?

A. 0.5600

B. 1.0056

C. 1.1800

D. 1.1856
